

Data Mining

Diplomatische Akademie Wien
Vienna School of International Studies
First Trimester 2025/2026
Course Type: Core
Course Credits: 4 ECTS

Lecture times: Please consult the online schedule. The lecture times and any possible changes are posted in the online schedule.
Office hours: Please see "Office hours" on the DA intranet
Convenor: Christoph Ivanusch
Christoph.ivanusch@gsi.uni-muenchen.de

Course Description

This course offers students a theoretical and practical introduction to data mining for the social sciences. With the transformative role of digitalisation in social life, vast amounts of observational data – such as behavioural, administrative, and textual data – are increasingly available for analysis. This development presents numerous opportunities, but also significant challenges for the social sciences. Traditional methods alone are often insufficient to make sense of such complex and high-volume data sets. Data mining equips social scientists with powerful tools to discover patterns, make predictions, and generate insights from unstructured or large-scale data. By learning these techniques, students will be better prepared to engage with contemporary research challenges and contribute to data-informed social inquiry.

Students will learn how to choose, apply, and reflect on data mining techniques for social science research. The course begins by addressing the selection and collection of large-scale data sets suitable for data mining. Next, students will learn how to prepare and preprocess these data sets (e.g., data cleaning and transformation) for meaningful analysis. The course then introduces a range of techniques for discovering patterns, making predictions, and generating insights, starting with regression, moving through traditional and deep learning approaches to classification, and concluding with unsupervised learning methods. Finally, the course will explore the role of open science in data mining, as well as current trends and developments, including large language models and the use of multimedia data.

The individual teaching blocks combine theoretical discussions with lab sessions, where students will apply key methods directly to data sets through practical exercises (using R). The course is designed to provide a solid foundation in data mining for the social sciences, offering both a basic theoretical understanding of core concepts and methods, and practical experience in applying them. This approach is intended to equip students with the skills needed to pursue their own data mining projects and to build on these competencies in later stages of their education or professional careers.

Course objectives:

- Selecting, collecting and preprocessing data: Students will learn how to select, gather and prepare data from various sources for analysis.
- Applying data mining techniques: Students will learn how to apply key methods – such as regression, classification (using both conventional techniques and deep learning), and unsupervised learning – for discovering patterns, making predictions, and generating insights from large-scale data sets.
- Reflecting techniques and findings: Students will learn how to critically assess the advantages, limitations, and implications of different data mining techniques in relation to their research findings.

Class prerequisites: Students are expected to have a basic understanding of statistics and programming (in R), particularly as applied to social science research.

Technical requirement: Students are required to bring their own laptops to class, with both R and RStudio pre-installed.

Assessments

Participation [15%]

Students are expected to actively participate in discussions and the practical lab sessions as well as to complete all required readings.

In-class exercises [15%]

Students are expected to complete all assigned in-class practical exercises and submit them on time.

Written examination [70%]

The written examination (120 minutes) consists of two applied tasks. Students are expected to draw on the techniques and methods covered in the course and implement them using R. To complete the exam, students must write appropriate code, include explanatory comments, and briefly reflect on the chosen techniques and their implications for social science research.

The written examination will take place in open-book format. Students may consult course materials, personal notes, and R documentation during the exam. However, all submitted work must be their own. Collaboration or use of external assistance (e.g., online forums, AI tools) is not permitted.

Examination

Final written examinations (denoted as the main assessment above) are scheduled in the timeframe of **December 5–12, 2025 (weekend and public holiday excluded)**.

Not being able to meet an assessment criteria must be documented by a **doctor's letter** to the Academic Administration. See section **Medical Leave** in the Student Handbook.

Re-take policy: Students can re-take the **main assessment** once in order to improve their performance dependent on the program. Re-takes have to be completed by the end of the next following trimester.

NB:

- Unexcused absence from examination/failure to submit a paper forfeits the second attempt.
- A paper without scientific/research content will not be accepted as a submission.
- For the award of a tuition waiver and/or participation in an exchange programme, only examination results for first attempts in core and advanced courses will be considered.
- The grade on the re-take examination/paper (second attempt) replaces the grade of the previously taken examination/submitted paper (first attempt) but contributes to the overall course assessment to the same extent. Having re-taken an examination/re-submitted a paper it is NOT possible to revert back to the initially awarded grade.
- A failing grade on a re-take examination/paper (second attempt) cannot be improved. The core/advanced/optional course may be repeated in the next following academic year (subject to availability). Both courses will be displayed on the transcript of records.
- Students who wish to re-take an examination/re-submit a paper for advanced/optional courses must send a request via email to paper-submission@da-vienna.ac.at immediately and no later than one week after the grade is entered into the student's records.
- For core courses, students who must re-take an examination/re-submit a paper are automatically registered for the second attempt at the end of the next following trimester.

Re-take examinations are scheduled in the week of March 9, 2026.

For further information, please check your [student account](#) and the Student Handbook (Chapter 9, "Regulations for Examinations, Seminar and Course Papers, Interviews").

Course Topics and Meetings

Block I. Foundations of data mining, data selection and collection

Session 1: Introduction

Overview of the course goals, expectations, structure and tools.

Session 2: What is data mining?

Conceptual overview, and discussion of relationship between conventional statistical methods and data mining.

Session 3: General strategies and key stages in data mining projects

Overview of data mining techniques; data mining pipeline: data selection and collection, preprocessing, modelling and evaluation.

Session 4: Data selection and collection (theoretical background)

Overview and discussion of important aspects of data selection and collection (sources, biases, sampling, methods, ethics).

Session 5: Data selection and collection (lab session)

Practical exercises in R on data collection.

Block II. Preprocessing and regression

Session 6: Preprocessing (theoretical background)

Overview and discussion of how to preprocess data (e.g., handle missing values, outliers, normalisation, transformations).

Session 7: Preprocessing (lab session)

Practical exercises in R on preprocessing.

Session 8: Regression (theoretical background)

Overview and discussion of (linear) regression methods (e.g., simple linear regression, multiple linear regression).

Session 9: Regression (lab session)

Practical exercises in R on (linear) regression.

Block III. Classification

Session 10: Classification – Conventional methods (theoretical background)

Overview and discussion of conventional classification methods (e.g., logistic regression, Naïve Bayes, tree-based models, SVM).

Session 11: Classification – Conventional methods (lab session)

Practical exercises in R on conventional classification methods.

Session 12: Classification – Deep learning (theoretical background)

Overview and discussion of deep learning neural networks for classification (e.g., single- and multi-layer neural networks, convolutional neural networks, recurrent neural networks).

Session 13: Classification – Deep learning (lab session)

Practical exercises in R on deep learning.

Block IV. Unsupervised learning, open science, current trends and developments

Session 14: Unsupervised learning (theoretical background)

Overview and discussion of principal components analysis, clustering and association analysis.

Session 15: Unsupervised learning (lab session)

Practical exercises in R on unsupervised learning.

Session 16: Open science and data mining

Discussion of open science in the field of data mining.

Session 17: Current trends and developments

Overview and discussion of current trends and developments in the field of data mining (e.g., transformer-based models and Large Language Models, multimedia data).

Session 18: Wrap-up and Q&A

Recap of key takeaways, discussion of (potential) applications by students, course feedback, and Q&A.

Readings

Fundamental readings

Fundamental readings are for orientation and for students lacking the preferred background knowledge for this course.

Douglas, A., Roos, D., Mancini, F., Couto, A., & Lusseau, D. (2022). *An introduction to R*. <https://intro2r.com>

Stauffer, R., Chimiak-Opoka, J., Rodríguez-R, L.-M., Simon, T., Zeileis, A. (2025). *Introduction to programming with R*. <https://discdown.org/rprogramming/>

Session 1: Introduction

Session 2: What is data mining?

Required readings:

Attewell, P., & Monaghan, D. (2015). Chapter 1: What is data mining? In *Data mining for the social sciences: An introduction* (pp. 3-12). University of California Press.

Attewell, P., & Monaghan, D. (2015). Chapter 2: Contrasts with the conventional statistical approach. In *Data mining for the social sciences: An introduction* (pp. 13-29). University of California Press.

Optional readings:

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 1: Introduction. In *An introduction to statistical learning: With applications in R* (pp. 1-14). Springer International Publishing.

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 2: Statistical learning. In *An introduction to statistical learning: With applications in R* (pp. 15-58). Springer International Publishing.

Session 3: General strategies and key stages in data mining projects

Required readings:

Attewell, P., & Monaghan, D. (2015). Chapter 4: Important stages in a data mining project. In *Data mining for the social sciences: An introduction* (pp. 53-62). University of California Press.

Optional readings:

Attewell, P., & Monaghan, D. (2015). Chapter 3: Some general strategies used in data mining. In *Data mining for the social sciences: An introduction* (pp. 30-52). University of California Press.

Sessions 4 and 5: Data selection and collection

Required readings:

Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Chapter 12: Scraping online data. In *Computational Analysis of Communication* (pp. 212-233). Wiley Blackwell.
<https://cssbook.net/>

Optional readings:

Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). Chapter 4: Selecting documents. In *Text as Data: A New Framework for Machine Learning and the Social Sciences* (pp. 41-47). Princeton University Press.

Sessions 6 and 7: Preprocessing

Required readings:

García, S., Luengo, J., Herrera, F. (2015). Chapter 3: Data Preparation Basic Models. In *Data Preprocessing in Data Mining* (pp 39–57). Springer International Publishing.

Sessions 8 and 9: Regression

Required readings:

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 3: Linear regression. In *An introduction to statistical learning: With applications in R* (pp. 59-128). Springer International Publishing.

Sessions 10 and 11: Classification – Conventional methods

Required readings:

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 4: Classification. In *An introduction to statistical learning: With applications in R* (pp. 129-196). Springer International Publishing.

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 8: Tree-based methods. In *An introduction to statistical learning: With applications in R* (pp. 327-366). Springer International Publishing.

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 9: Support vector machines. In *An introduction to statistical learning: With applications in R* (pp. 367-402). Springer International Publishing.

Optional readings:

Attewell, P., & Monaghan, D. (2015). Chapter 9: Classifiers. In *Data mining for the social sciences: An introduction* (pp. 133-161). University of California Press.

Attewell, P., & Monaghan, D. (2015). Chapter 10: Classification trees. In *Data mining for the social sciences: An introduction* (pp. 162-184). University of California Press.

Sessions 12 and 13: Classification – Deep learning

Required readings:

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 10: Deep learning. In *An introduction to statistical learning: With applications in R* (pp. 403-460). Springer International Publishing.

Optional readings:

Attewell, P., & Monaghan, D. (2015). Chapter 11: Neural networks. In *Data mining for the social sciences: An introduction* (pp. 185-195). University of California Press.

Sessions 14 and 15: Unsupervised learning

Required readings:

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). Chapter 12: Unsupervised learning. In *An introduction to statistical learning: With applications in R* (pp. 495-550). Springer International Publishing.

Optional readings:

Attewell, P., & Monaghan, D. (2015). Chapter 12: Clustering. In *Data mining for the social sciences: An introduction* (pp. 196-215). University of California Press.

Session 16: Open science and data mining

Required readings:

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., ... & Mons, B. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific data*, 3(1), 1-9.

Session 17: Current trends and developments

Required readings:

Alizadeh, M., Kubli, M., Samei, Z., Dehghani, S., Zahedivafa, M., Bermeo, J. D., Korobeynikova, M., & Gilardi, F. (2025). Open-source LLMs for Text Annotation: A Practical Guide for Model Setting and Fine-tuning. *Journal of Computational Social Science*, 8(1), 1-25.

Van Atteveldt, W., Trilling, D., & Calderón, C. A. (2022). Chapter 14: Multimedia data. In *Computational Analysis of Communication* (pp. 258-282). Wiley Blackwell.
<https://cssbook.net/>

Session 18: Wrap-up and Q&A